

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

Paper 1 Multiple Choice

9744/01

25 September 2018

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name and CT on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

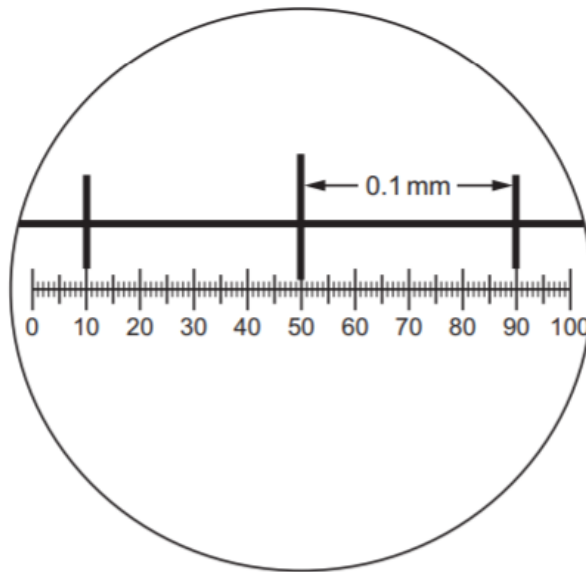
Any rough working should be done in this booklet.

Calculators may be used.

This document consists of **17** printed pages and **1** blank page.

[Turn over

- 1 The diagram shows a stage micrometer scale viewed with an eyepiece graticule, using a magnification of $\times 200$.



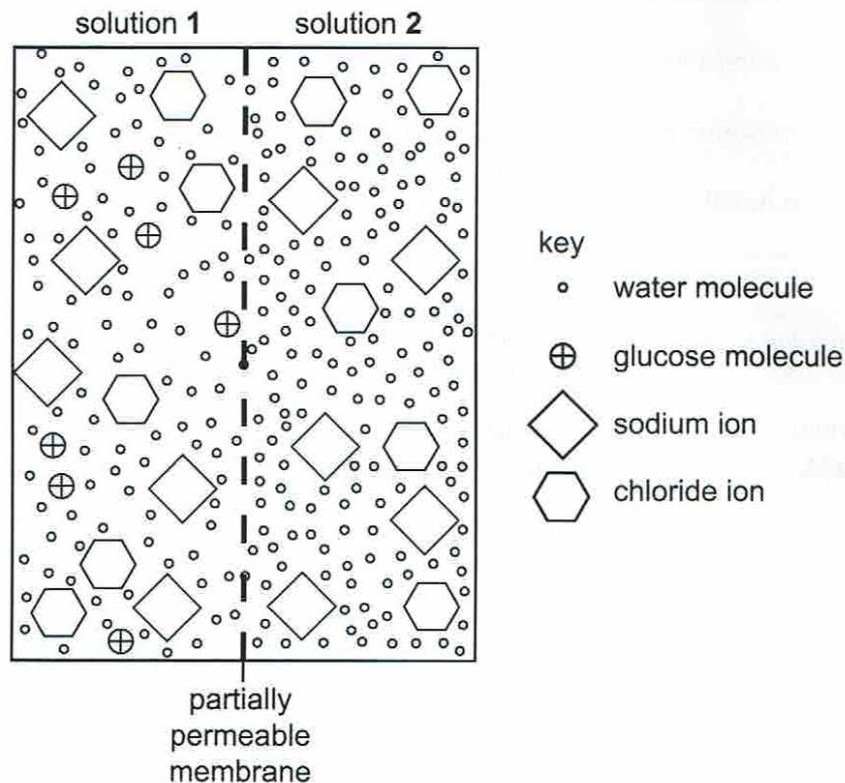
Using the same magnification, a chloroplast is measured as 4 eyepiece graticule divisions long.

How long is the chloroplast?

- A $1.0 \times 10^1 \mu\text{m}$ B $4.0 \times 10^2 \mu\text{m}$ C $2.5 \times 10^{-1} \mu\text{m}$ D $2.5 \times 10^{-2} \mu\text{m}$
- 2 Which of these processes allow movement in both directions across cell surface membranes?
- 1 active transport
 - 2 diffusion
 - 3 facilitated diffusion
 - 4 osmosis
- A 1, 2, 3 and 4 B 1 and 4 only C 2 and 3 only D 2 only

- 3 The diagram represent a partially permeable membrane separating two solutions, **1** and **2**.

The relative diameters and concentrations of the ions and molecules in the solutions and the pore diameter of the partially permeable membrane are drawn proportionately in the diagram.

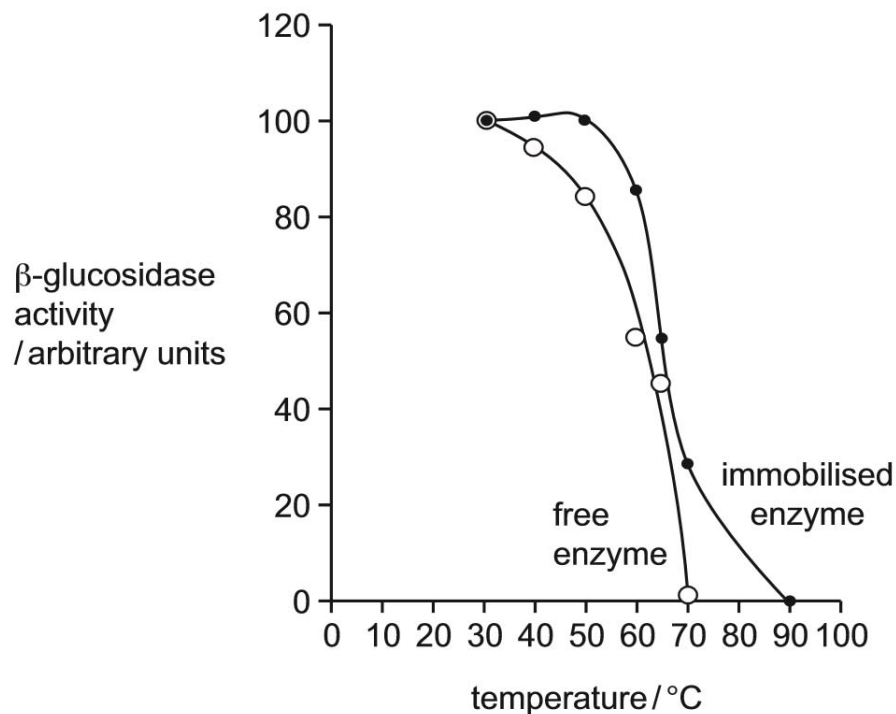


Which statement about the movement of the molecules and ions between solution **1** and solution **2** is correct?

- A** Net movement of water occurs down a water potential gradient from solution **2** to solution **1**, movement of glucose occurs down a diffusion gradient from solution **1** to solution **2**, and no movement of sodium and chloride ions occurs across the partially permeable membrane.
- B** Sodium and chloride ions pass through the partially permeable membrane equally in both directions because they are of the same concentration in solution **1** and solution **2**, but glucose and water diffuse in opposite directions until concentration of solution **1** and solution **2** are equal.
- C** Solution **1** has a lower water potential than solution **2** causing net movement of water from solution **2** to solution **1** until the water potentials are the same; all the solutes are too large to pass through the membrane.
- D** The difference in water potential causes water to move by osmosis from solution **1** to solution **2**, a diffusion gradient causes net movement of glucose from solution **1** to solution **2**, and sodium and chloride ions do not move between the two solutions.

- 4 An experiment was conducted to investigate the effect of temperature on the activity of the enzyme β -glucosidase. The enzyme was tested when in solution (free) and when immobilised in alginate beads.

The results are shown in the graph below.



Which statement about the effect of immobilisation of β -glucosidase is correct?

- A It increases the kinetic energy of the enzyme.
 - B It inhibits the activity of the enzyme.
 - C It reduces the optimum temperature of the enzyme.
 - D It stabilizes the enzyme against denaturation.
- 5 Which statement about the consequences of producing genetically identical cells is **not** correct?
- A All cells will have the same phenotypes.
 - B All diploid cells will have the same alleles at the same loci.
 - C All genes will be passed to the daughter cells.
 - D All the coding sections of DNA will be preserved.

6 Which statements refer to roles of meiosis in humans?

- 1 It reduces the number of chromosomes.
- 2 It leads to the production of gametes.
- 3 It results in cells that contain identical sets of chromosomes.
- 4 It ensures that a fertilized cell has the diploid number of chromosomes.

- A 1, 2, 3 and 4
B 1, 2 and 4 only
C 1 and 3 only
D 2 and 4 only

7 Which molecules are found in both the cytoplasm and the nucleus of a typical eukaryotic cell?

- 1 DNA nucleotides
- 2 DNA polymerase
- 3 RNA nucleotides
- 4 RNA polymerase

- A 1, 2, 3 and 4 B 1, 2 and 4 only C 2, 3 and 4 only D 3 and 4 only

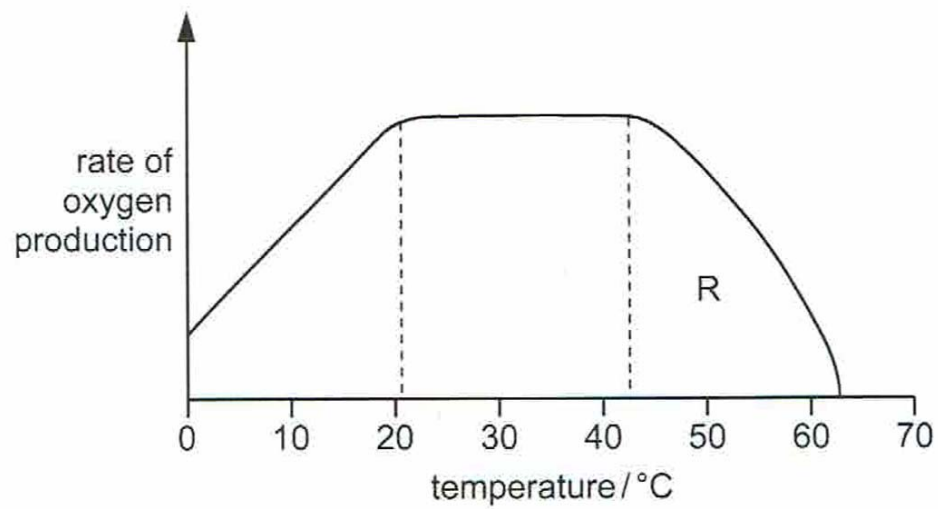
8 Francis Crick and James Watson, who published the first acceptable structure for DNA, added this statement to their account.

“It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material.”

What is the specific pairing they mention for the copying mechanism now known to take place?

- A Each purine with either pyrimidine and each pyrimidine with either purine
B Each purine with itself or the other purine and each pyrimidine with itself or the other pyrimidine
C Each pyrimidine with a particular purine and each purine with a particular pyrimidine
D Each pyrimidine with itself and each purine with itself

- 9 The graph shows how the rate of oxygen production by the photosynthetic alga, *Chlamydomonas* varies with temperature, when all other factors are kept constant.

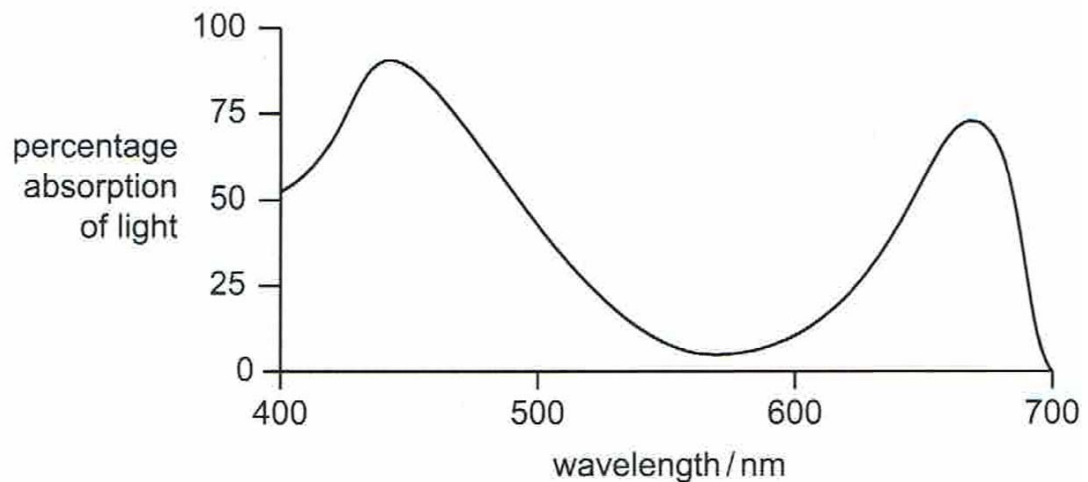


The following are factors that can limit the rate of reaction.

- 1 Carbon dioxide concentration
- 2 Light intensity
- 3 temperature

A 1, 2, and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 3 only

- 10 The graph shows the absorption of light at different wavelength by intact chloroplasts from a pond weed.



A sample of the same pond weed was exposed to four different wavelengths of light of the same intensity for the same time. The table shows the number of bubbles produced by the pond weed at each wavelength of light.

experiment	number of bubbles			mean number of bubbles
1	15	14	16	15
2	12	11	13	12
3	3	4	2	3
4	1	2	0	1

Which row shows the number of bubbles produced by the different wavelengths of light investigated?

	mean number of bubbles			
	440 nm	520 nm	560 nm	670 nm
A	1	12	15	3
B	3	1	12	15
C	12	15	3	1
D	15	3	1	12

- 11** Labrador Retrievers, a popular breed of dog, can have several possible coat colours – black, chocolate or yellow.

Individuals with the genotype BBEE are black, while individuals with the genotype bbee are yellow. A cross between individuals with genotypes BBEE and bbee resulted in offspring that were all black in colour.

The F1 offspring were sibling mated, and the coat colours of the resulting F2 offspring were recorded.

<u>Coat colour</u>	<u>No.</u>
Black coat	565
Chocolate coat	184
Yellow coat	254

Which of the following options correctly describes the mode of inheritance of coat colour in Labrador Retrievers?

- A** Collaboration **B** Co-dominance **C** Sex linkage **D** Epistasis

- 12** The *t*-test (two tailed) value for an experiment with total sample size of 16 was found to be 2.100.

	SIGNIFICANCE LEVEL FOR TWO-TAILED TEST				
df	0.20	0.10	0.05	0.02	0.01
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921

Which of the following combinations shows the correct conclusion?

	Degree of freedom	Probability	Conclusion
A	16	Between 0.05 and 0.10	Do not reject the null hypothesis
B	16	Between 0.02 and 0.05	Reject the null hypothesis
C	14	Between 0.05 and 0.10	Do not reject the null hypothesis
D	14	Between 0.02 and 0.05	Reject the null hypothesis

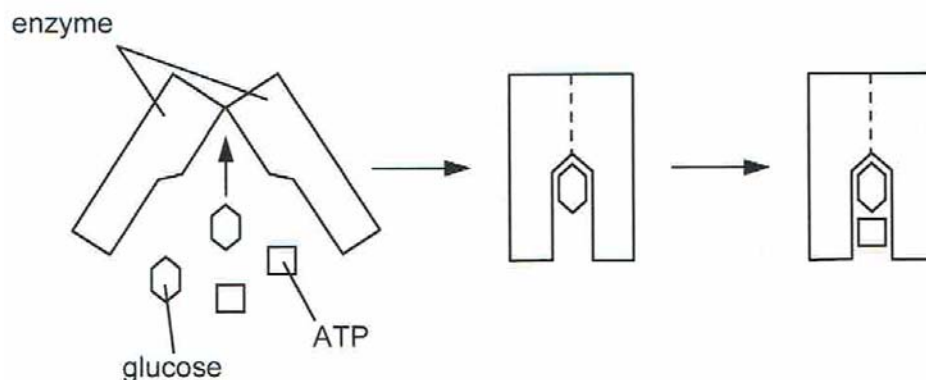
13 Which row shows where some stages of aerobic respiration occur in a eukaryotic cell?

	Link reaction	Oxidative phosphorylation
A	Inner mitochondrial membrane	Mitochondrial intermembrane space
B	mitochondrial matrix	Inner mitochondrial membrane
C	mitochondrial matrix	mitochondrial matrix
D	Outer mitochondrial membrane	Inner mitochondrial membrane

14 Which set of reactions release the largest number of ATP molecules from one molecule of glucose?

- A** conversion of glucose to carbon dioxide and ethanol
- B** conversion of glucose to carbon dioxide and water
- C** conversion of glucose to lactic acid
- D** conversion of glucose to pyruvate

15 The enzymes, hexokinase, which catalyses the transfer of a phosphate group from ATP to glucose during glycolysis, changes shape during the reaction, as shown in the diagram.

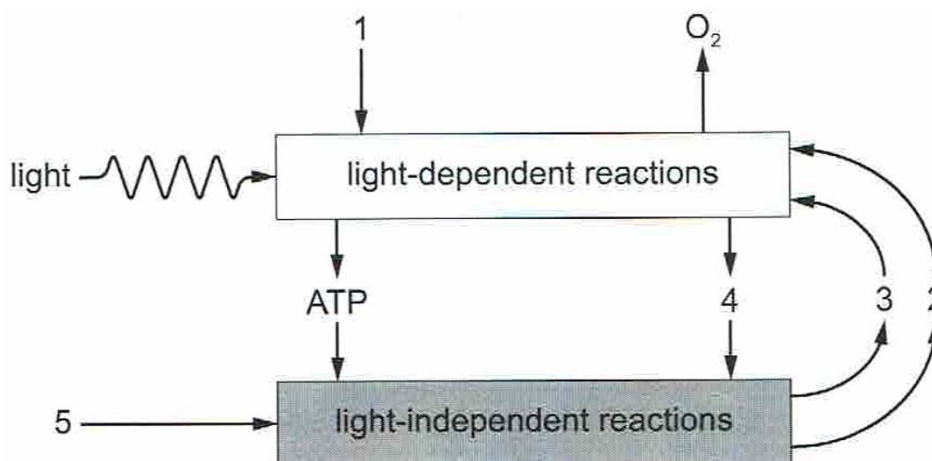


What statement describe the behaviour of this enzyme?

- 1** The reaction illustrates the 'lock and key' hypothesis of enzyme action.
- 2** Binding glucose to the enzyme allows ATP to bind also.
- 3** The active site for ATP is formed only in the presence of glucose.
- 4** In the absence of glucose, ATP is not hydrolysed to release a phosphate group.

- A** 1, 2 and 3 **B** 1, 2 and 4 only **C** 2, 3 and 4 only **D** 3 and 4 only

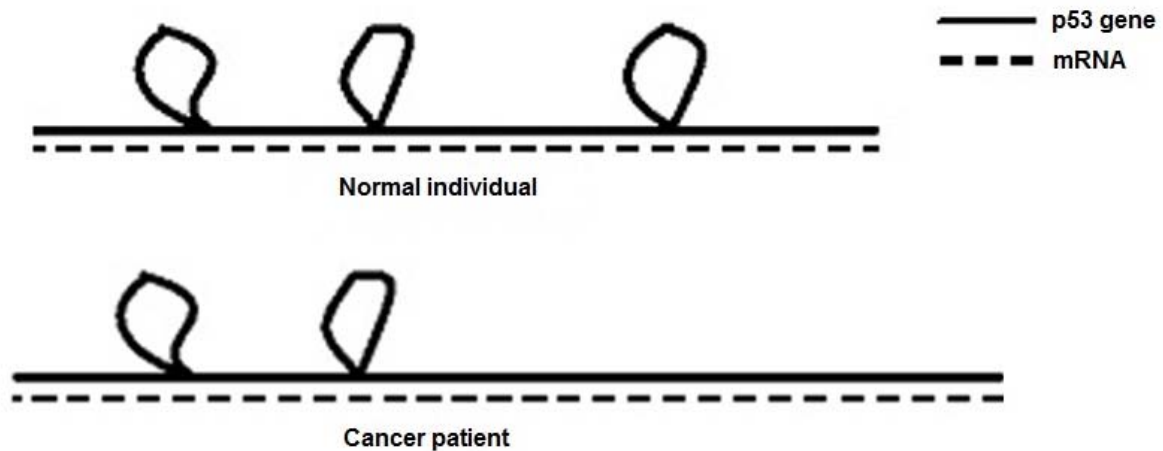
16 The diagram summaries the process of photosynthesis.



Which row identifies the reactants 1, 2, 3, 4 and 5?

	1	2	3	4	4
A	Carbon dioxide	ADP + Phosphate	reduced NAD	NAD	water
B	Carbon dioxide	reduced NADP	ADP + Phosphate	NADP	water
C	water	NAD	reduced NAD	ADP + Phosphate	Carbon dioxide
D	water	NADP	ADP + Phosphate	reduced NADP	Carbon dioxide

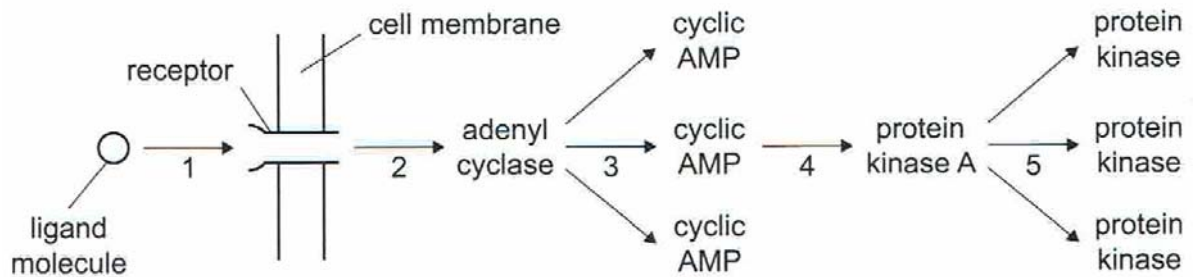
- 17 mRNA was isolated from a normal individual and a patient suffering from cancer. The mRNA was allowed to hybridise with the *p53* gene. The schematic diagram shows the results of the hybridisation process under the electron microscope



Which of the following could be a possible explanation why the patient is suffering from cancer?

- A A point mutation had occurred in the intron leading to the failure to excise one intron, hence leading to a longer dysfunctional protein being translated.
 - B A point mutation had occurred in the intron leading to an exon being excised, hence leading to a shorter dysfunctional protein being translated.
 - C A point mutation had occurred leading to the failure of spliceosome to recognise splice sites leading to the excision of the wrong intron, leading to a dysfunctional protein being translated.
 - D Gene amplification had occurred leading to the multiple copies of a trinucleotide repeat in an intron, hence causing splice site to be misread due to frameshift mutation, leading to a longer dysfunctional protein being translated.
- 18 Cancer is caused by changes in the genes which control cell division. Which of the following changes would not result in uncontrolled cell division?
- A Tumour suppressor genes become less active.
 - B Stimulation of cell division by platelet-derived growth factor.
 - C Absence of contact inhibition.
 - D Infection by certain viruses, e.g. hepatitis B which carry oncogenes.

19 The diagram shows an example of cell signaling.



At how many of the five numbered stages does signal amplification occur in this example?

- A 1 B 2 C 3 D 4

20 Which of the following statements could explain why a combination of different drugs rather than a single drug is being used to treat HIV patients?

- (i) HIV has a short generation span (eg. 2 days).
- (ii) In an AIDS patient, the HIV infection produces many new viruses per day (eg. 10^{10} new viruses per day or more).
- (iii) HIV has an RNA genome which has a higher mutation rate than DNA as it is single stranded.
- (iv) The RNA genome allows the HIV to mutate rapidly to acquire the resistance to the drug being used.
- (v) Insertion of the HIV genome into the host cells will result in the host cells mutations that will confer resistance to the drug.

- A (i) And (ii) only
 B (iii) and (v) only
 C (i), (ii) and (iii) only
 D (iii), (iv) and (v) only

21 Two mutant strains of *Escherichia coli*, one having F plasmid and the other without the F plasmid, were mixed in suspension. One strain was unable to synthesise the amino acid threonine, the other was unable to manufacture the two amino acids leucine and lysine. The genes coding for the synthesis of these essential metabolites are widely separated on the bacterial chromosome.

After sixty minutes, the bacteria were plated onto a growth medium deficient in all three metabolites. Four hundred colonies grew.

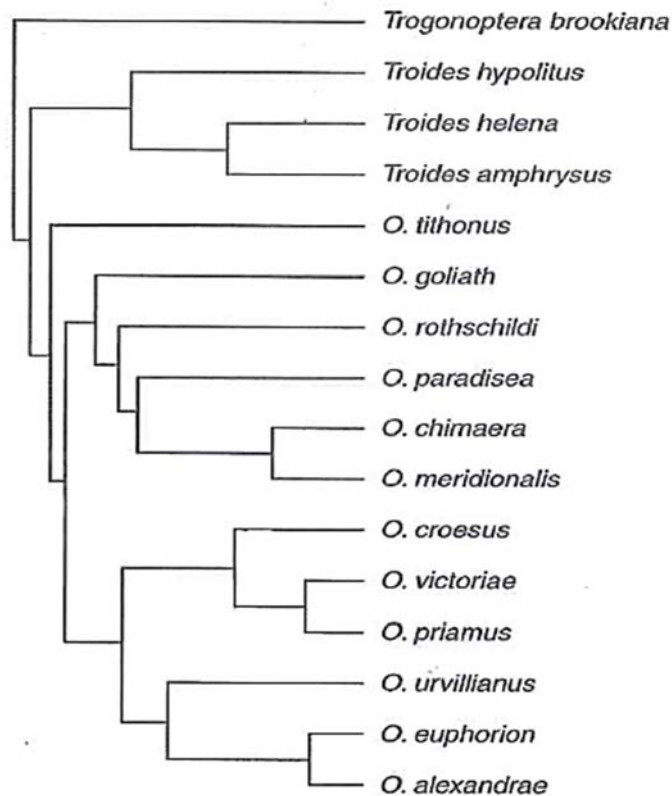
Which process could explain the growth of these colonies?

- A conjugation B translocation C transduction D transformation

22 Which statement defines all control elements?

- A A segment of DNA to which RNA polymerase binds preferentially.
- B A short region of DNA that can bind with proteins to enhance transcription levels.
- C DNA sequences that interact with proteins to determine the rate and timing of gene expression.
- D Proteins found only in eukaryotes that bind to DNA sequences to control transcription.

23 The figure is a phylogenetic tree of three genera of butterflies (*Orniithoptera*, *Trogonoptera* and *Troides*) that was constructed based on the comparison of the nucleotide sequences of the gene *ND5* that is located in the mitochondrial genome.



Based on the phylogenetic tree, what conclusions can be drawn regarding the relationships of these three genera?

- 1 The three genera *Orniithoptera*, *Trogonoptera* and *Troides* form a monophyletic clade.
- 2 *O. victoriae* shares fewer identical nucleotides in the *ND5* gene with *O. alexandrae* than with *O. goliath*.
- 3 *Troides hypolitus* shares both ancestral and shared derived traits with *Troides helena* and *Troides amphrysus*.
- 4 *Trogonoptera brookiana* diverged from the common ancestor much earlier than *O. alexandrae* so it is now extinct.

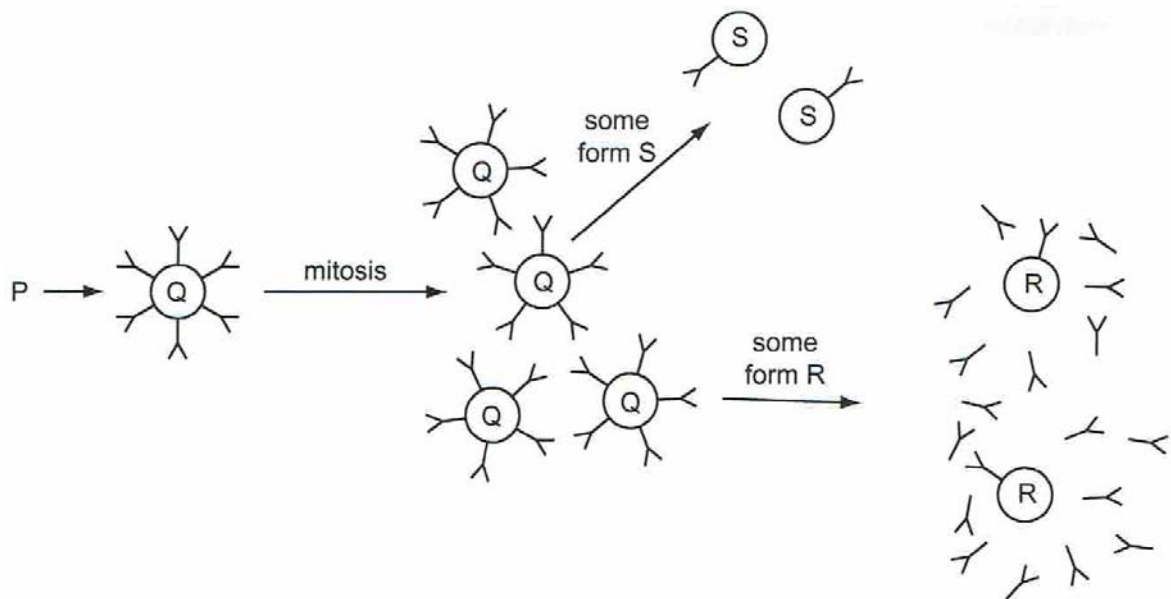
- A 1 only B 1 and 3 only C 2 and 4 only D 2, 3 and 4 only

- 24** A biologist discovers two populations of wolf spiders whose members appear identical. Members of one population are found in the leaf litter deep within the woods. Members of the other population are found in the grass at the edge of the woods. The biologist decides to designate the members of the two populations as two separate species

Which species concept has this biologist used and what is its limitation of this concept?

	Species concept	Limitation
A	Morphological species concept	Similarities in structures might have arisen due to convergent evolution.
B	Morphological species concept	Cannot be used to group fossils or organisms which are completely asexual in their reproduction.
C	Ecological species concept	Difficult to determine the magnitude of genetic variation required to distinguish between 2 putative species.
D	Ecological species concept	Difficult to determine what is considered as different niches, especially when organisms use resources from another niche during time of scarcity.

- 25** The diagram shows part of an immune response.

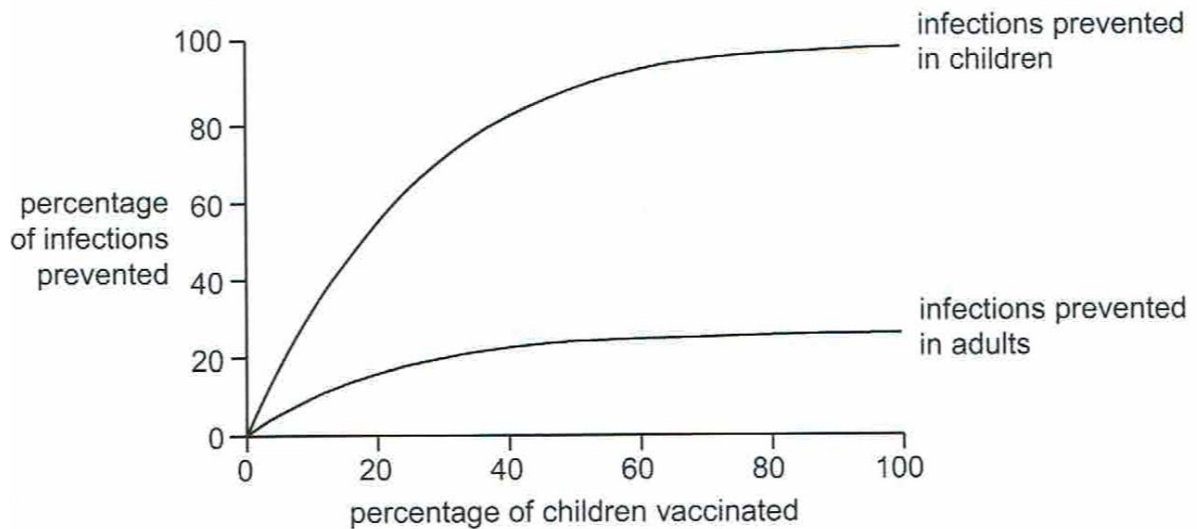


Which row correctly identifies P, Q, R and S?

	P	Q	R	S
A	antigen	T-lymphocyte	memory cell	B-lymphocyte
B	APC	B-lymphocyte	plasma cell	memory cell
C	pathogen	T-lymphocyte	B-lymphocyte	plasma cell
D	T-lymphocyte	B-lymphocyte	plasma cell	T-lymphocyte

26 When sufficient individuals are vaccinated, the disease transmission cycle can be broken.

The diagram shows the effect of vaccination of children on the prevention of infection.



What can be concluded about the effect of vaccination of children from this data?

- 1** When approximately 80% of children are vaccinated, the cycle of disease transmission in children is broken.
- 2** Vaccination of children reduced the percentage of infections in both adults and children.
- 3** The effect on adult infection is less than that on infection in children, because adults will have been vaccinated as children.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

27 What is the role of stem cells with regards to the function of adult tissues and organs?

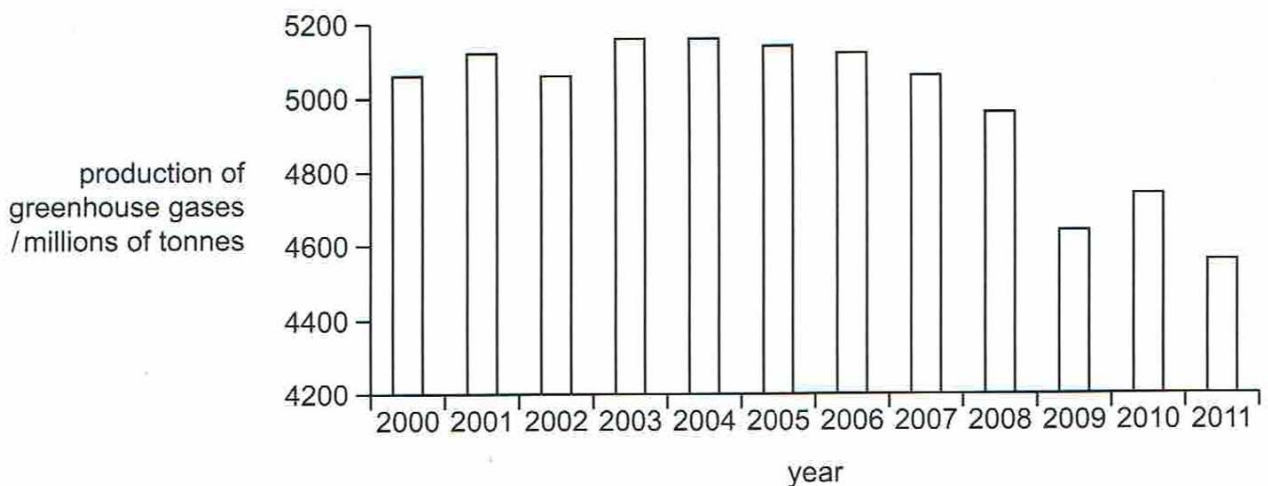
- A** Stem cells are fully differentiated cells that reside under the surface of epithelial tissue, in position to take over the function of the tissue when the overlying cells become damaged or worn out.
- B** Stem cells are totipotent cells that divide asymmetrically, giving rise to one daughter cell that remains a stem cell and one daughter cell that will differentiate to replace damaged and worn out cells in the adult tissue or organ.
- C** Stem cells are embryonic cells that persist in the adult, and can give rise to all of the cell types in the body.
- D** Stem cells are cells that have yet to express the genes and produce proteins characteristic of their differentiated state, but do so when needed for repair of tissues and organs.

- 28** Stem cells are found in many tissues that require frequent cell replacement, such as the skin and the blood.

A bone marrow stem cell that is transferred to the skin is never induced to produce a skin cell, and a skin stem cell that is transferred to the bone marrow is never induced to produce a blood cell.

Which statement explains this?

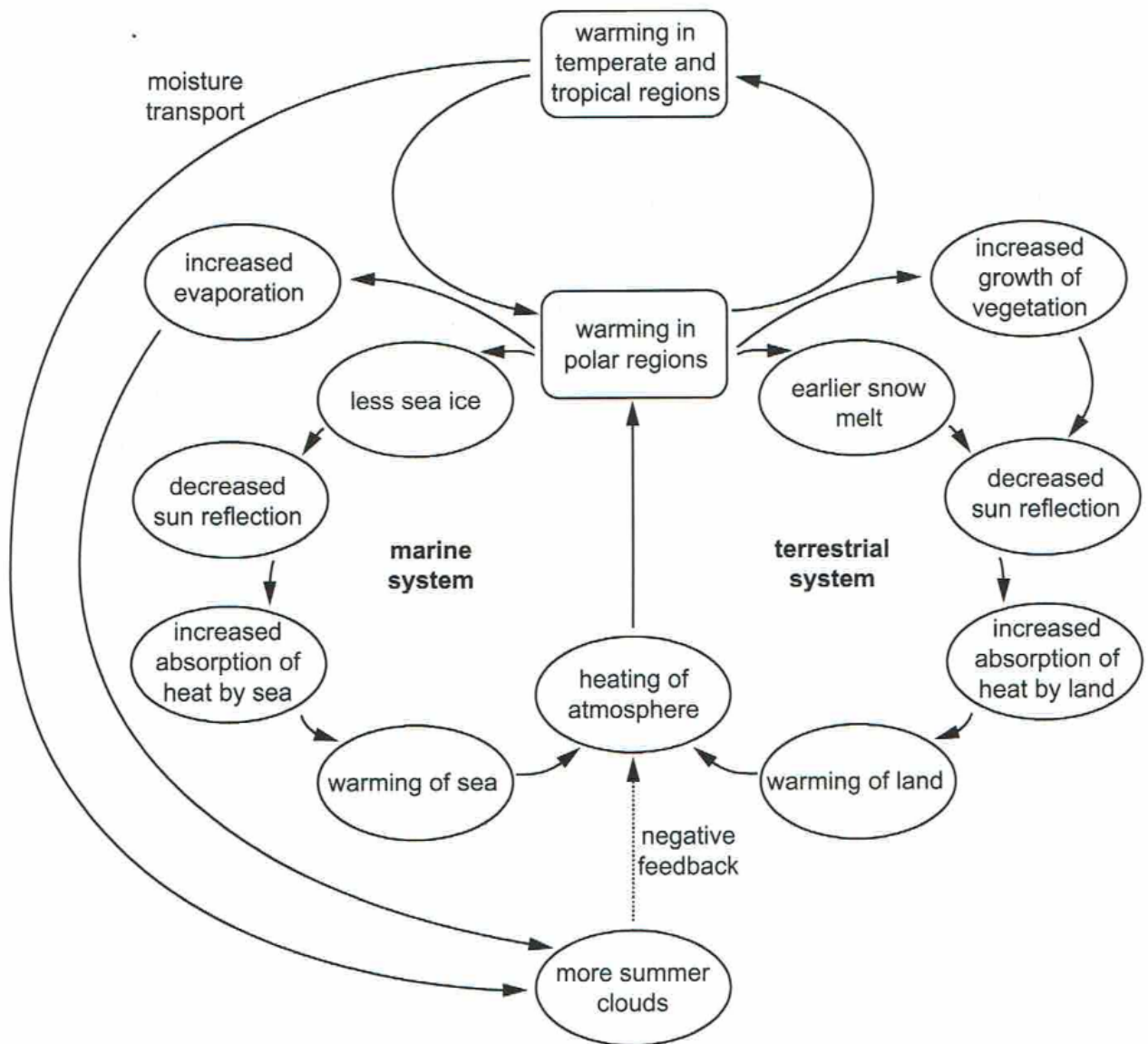
- A** Binding of repressor molecules prevents the expression of genes not required for a particular cell line.
 - B** Different stem cells have only the genes required for their particular cell line.
 - C** Expression of genes not required for a particular cell line is controlled at translational level.
 - D** Genes not required for the differentiation of a particular cell line are methylated.
- 29** The bar chart shows the production of greenhouse gases (carbon dioxide and methane) from agriculture in the European Union (EU) from 2000 to 2011, measured in millions of tonnes.



Which of the following could contribute to the trend seen between 2003 and 2009?

- A** Conversion of intensive farmland into woodland reserves
- B** Greater use of agricultural machinery for harvesting
- C** Increased consumption of meat-based products
- D** Increased import and export of crops between EU countries.

- 30 The diagram shows the effect of increasing temperatures on the ice and snow cover at the polar regions.



Which effect of higher temperatures on the polar regions could increase global warming?

- A Increased evaporation leads to more rainfall, which absorbs heat from the land and sea.
- B Melting of ice and snow results in less reflection of sunlight and more heat absorption by the Earth.
- C Melting of sea ice causes more cloud formation, which increases absorption of heat in the atmosphere.
- D Earlier melting of snow allows vegetation cover to increase faster, reducing loss of heat from the surface of the Earth.